Craig Fouts

CRAIGFOUTS.COM
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CORE SKILLS

Python | C | C++ | PyTorch | Pyro | Scikit-Learn | SciPy | ScanPy | AnnData | Muon | Sphinx | SDL3 | Probabilistic Modeling | Bayesian Inference | Interpretable Machine Learning | System Identification | Mechanistic Modeling | Functional Genomics | Spatial Proteomics

EDUCATION

Columbia University | **MSc**
Applied Mathematics **2022 – 2023**

Completed a graduate research assistantship in the New York Genome Center Technology Innovation Laboratory.

The Ohio State University | **BSc**
Computer Science/Mathematics **2018 – 2022**

Received honors research distinction for original research and publication in collaboration with Google Research.

EXPERIENCE

Uppsala University | **Vicković Group**, **Science for Life Laboratory**
Computational Research Engineer **Oct 2024 – Sep 2025**

Designed and developed a nonparametric generative model called **ATLAS** that elucidates anatomical niches and pathological structures in spatial-omics datasets based on proteomic or transcriptomic profile and spatial distribution. The model adaptively learns the number of spatially-resolved niches while clarifying shared motifs across histological batches.

New York Genome Center | **Technology Innovation Laboratory**
Associate Computational Biologist II **Jan 2024 – Sep 2024**
Graduate Research Assistant **Oct 2022 – Dec 2023**

Developed a probabilistic spatial topic model called **sceLDA** that clusters anatomical niches in spatial transcriptomics datasets based on cell type composition and spatial distribution. The model is part of **Hudson**, a spatial-omics ecosystem for repurposing decommissioned Illumina HiSeq 2500 sequencers as accessible immunofluorescent imaging platforms.

Mentored an undergraduate intern who successfully derived and implemented a coordinate ascent variational inference algorithm for a statistical workflow used to explore how epigenetic states influence gene expression.

The Ohio State University | **Translational Data Analytics Institute**
Undergraduate Research Assistant **Aug 2021 – Sep 2022**

Developed a computational pipeline for aggregating, annotating, and analyzing multi-modal (CO2, PM2.5/10, noise, etc.) data compiled from environmental sensors used to study the long-term effects of aircraft engines in urban neighborhoods. The pipeline uses dynamic time warping to align several data streams based on location and ambient conditions.

PUBLICATIONS

Growing Steerable Neural Cellular Automata
Ettore Randazzo, Alexander Mordvintsev, **Craig Fouts** **2023** | **ALIFE**

Angle- and gradient-based steerable adaptations of neural cellular automata (NCA) for investigating morphogenetic chirality and out-of-training cell dynamics with structured seeds and per-channel Sobel perception filters. This work explores how directed initial conditions can be used to drive the model toward targeted stable states without explicit training.

https://doi.org/10.1162/isal_a_00564

Growing Isotropic Neural Cellular Automata

Alexander Mordvintsev, Ettore Randazzo, **Craig Fouts** **2022 | ALIFE**

A modified fully isotropic extension of neural cellular automata (NCA) for modeling orientation-agnostic morphogenesis via structured seeds, a per-channel Laplacian perception filter, and a rotation-invariant loss function. The model parameterizes a nonlinear recurrence relation using a convolutional neural network with residual connections for each step.

https://doi.org/10.1162/isal_a_00552

PORTFOLIO

Fitting ODEs with NumPyro

An introductory guide on building ordinary differential equation (ODE) models using NumPyro's implementation of Hamiltonian Monte Carlo (HMC) inference with adaptive No U-Turn Sampling (NUTS). The guide walks through setting up an autonomous one-dimensional ODE model, fitting the model to generated data, and visualizing parameter estimations.

https://www.craigfouts.com/tutorials/numpyro/

Asterism: Experiments with Point Cloud Clustering

A compilation of semantic segmentation models and dimension reduction algorithms implemented in Python/C and adapted for point cloud data. Current methods include variations of Latent Dirichlet Allocation (LDA), several neural topic models, and a neural clustering process (NCP). This repository will soon be available as an open-source Python package.

https://www.craigfouts.com/asterism/

ACCOLADES

Honors

The Ohio State University: Magna Cum Laude | Honors Research Distinction **2022**
Granville High School: Cum Laude Society | National Honor Society | Sociedad Honoraria Hispánica **2017**
Scouting America: Eagle Scout **2016**

Competitions

HackOHI/O Hackathon: 1st Place Grand Prize | Microsoft Challenge Winner | People's Choice Award **2021**
Ohio State FEH Honors Robotics Competition: 2nd Place Outstanding Achievement in Innovation **2019**
OMEA Solo & Ensemble: Rank 1 Class A Violin Solo Performance **2016** & **2017**

COURSEWORK

Columbia University

Applied Statistics III (A) | Machine Learning for Functional Genomics (A) | Advanced Linear Algebra (A+) **2023**
Numerical Algebra & Optimization (A) | Partial Differential Equations (A-) **2022**

The Ohio State University

Discrete Mathematical Models (A) | Quantitative Neuroscience (A) | Computer Networking (A) **2022**
Mathematical Statistics II (A) | Advanced Artificial Intelligence (A) | Programming Languages (A) **2021**
Data Structures & Algorithms (A) | Experimental Physics (A) | Intermediate Mechanics (A-) **2020**
Ordinary Differential Equations (A) | Honors Physics Electricity & Magnetism (A) **2019**
Honors Real Analysis (A) | Honors Psychology (A) **2018**